

Nordic ID AR85

Nordic ID AR85 is a powerful fixed area UHF RFID beam reader with a sophisticated phased-array antenna covering up to 120 m² of floor space. With reading speeds up to 1000 tags/s and dual-linear polarization, it efficiently monitors designated areas and detects hard-to-read tags.



UHF RFID

Supported standard	ISO 18000-63 (EPC Class 1 Gen 2v2)
Frequency band	ETSI 865.6–867.6 MHz or FCC/IC 902–928 MHz
Regulatory	CE ETSI EN 302 208, CE ETSI EN 301 489, FCC part 15.247 IC RSS-210, Safety IEC 62368-1:2019 ed. 3
Max receive sensitivity	-81 dBm
Typical reading speed	Up to 1000 tags/s
Radiated power	33 dBm (2 W) ERP / 4 W EIRP
Antenna features	13 radiation beams (26 states with dual polarization), Dual-Linear, 3 dB beam width 45°, Beam steering ±40
Floor coverage area per reader	Max. 120 m ² (diameter = 2 x installation height)
Conducted radiated power for external antenna ports	30 dBm (1 W)
External antenna port	2 pcs. 50 Ω / SMA Female

PLATFORM

CPU	1.2GHz 64-bit Quad-Core Cortex-A53 processor
Operating system	Debian based embedded Linux
Memory	1 GB RAM / 4 GB flash

USER INTERFACE

Indicators	4 pcs (programmable) leds
SIM	Mini-SIM

CONNECTIVITY

USB	USB host, type A / USB device, type B, USB HID class supported
GPIO	4 inputs / 4 outputs, opto isolated
Wireless WAN (optional)	3G HSPA / 2G GPRS / EDGE
LAN	Ethernet 10/100 Mbit
Wireless LAN (optional)	IEEE 802.11 a/b/g/n, dual band 2.4 / 5 GHz

POWER

External Power Supply	PoE 802.3af or AC/DC adapter: 100-240 VAC / 24 VDC, 1.25A
Operating Power	12W PoE, 20 W DC

SIZE AND WEIGHT

Dimensions	(W) 405 x (L) 405 x (H) 68 mm
Weight	2.4 kg / 5.3 lb

ENVIRONMENT

Environmental sealing	IP20, for indoor use only
Operating Temperature	-20 °C to 55 °C
Storage Temperature	-40 °C to 85 °C
Mounting	VESA 75/100 Compatible

SOFTWARE INTERFACE

Data management	Nordic ID Radea
Device management	Nordic ID Radea, SOTI®
Firmware update	Via Web management UI and RESTful service
API support	NUR API for RFID and RESTful service
Other protocols	LLRP

FEATURE HIGHLIGHTS

- Smart fixed area reader for automated UHF RFID data collection
- Enables real-time visibility and location/movement detection
- Integrated Linux based computer for versatile SW development
- Cutting edge UHF RFID sensitivity for reliable detection
- Improved to minimize interferences of close-by readers
- Can be utilized as a WLAN access point

PRODUCT GROUP

Category	RFID Reader
Product Group	Fixed Area UHF RFID Beam Reader